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On modeling nonlinear phenomena under the influence of an electromagnetic field in a dense plasma

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Abstract – The quadratic response function (QRF) of a dense plasma in a constant magnetic field to an electromagnetic field is considered. To determine this function, a self-consistent model approach was applied based on the use of the quadratic fluctuation-dissipative theorem (FDT), frequency moments for the quadratic dense plasma reaction function, and explicit approximation for QRF with a fitting parameter. The conditions of using QRF for the interpretation of laboratory experiments on the study of nonlinear phenomena caused by the quadratic interaction of electromagnetic waves in a dense charged medium (Coulomb systems, plasma) in a constant magnetic field are discussed.

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Introduction. – Nonlinear phenomena in various media (crystals, graphene, various mixtures, composites, weakly ionized gases, suspensions, etc.) related to the nonlinear interaction of electromagnetic waves have been intensively studied in recent works (see [1–3] and references therein). At the same time, nonlinear phenomena in dense charged systems, *i.e.*, media with strong inter-particle interaction, Coulomb systems and plasma (when the energy of interaction of the particles of the medium with each other per particle $-U$, is comparable to the average kinetic energy of the particle $-k_B T$), have not been studied. Therefore, firstly in this paper phenomena were caused by a quadratic reaction to an electromagnetic field: second-harmonic generation (SHG), parametric generation of radiation (PGR) are considered in this media. In dense plasma, SHG and PGR could be implemented in a laboratory setting in the presence of stationary relatively strong magnetic field. Calculated characteristics of these phenomena can be determined by the quadratic response function of a charged media in a constant magnetic field to an electromagnetic field.

As studies show (see, for example, [1–3], and references therein), it is impossible to consistently calculate the QRF of dense charged media to perturbations within the framework of nonlinear response theory, computer modeling of these functions has not been performed. Therefore, it is

advisable to use model approaches to determine the functions. One of the variants of the model may be the use of explicit approximation for QRF with fitting parameters that are found from the exact frequency moments of such functions. In this paper, based on the results of calculating the frequency moments of the QRF [2,3], the structure of the QRF to an electromagnetic field of a dense laboratory electron-ion hydrogen ($n_e = n_i$) plasma in a constant magnetic field is considered using the above-mentioned model. The results of the proposed self-consistent QRF model of laboratory dense plasma in a constant magnetic field are used to determine the conditions for conducting possible experiments to study PGR and SHG in such an environment based on approaches used in nonlinear optics (see the following two sections). A comparison was made with the previously obtained our data on PGR and SHG simulation [1,3].

The quadratic fluctuation-dissipation theorem. –

In this section, a quadratic fluctuation-dissipation theorem for tensor and scalar response functions is formulated. At the same time, the basic designations, ratios, and definitions of QFR and correlators, obtained within the framework of response theory and presented below in a concise form, are discussed in detail and are available in [2,3].

Let us write out a quadratic FDT (see [2,3]), on which the approach used in the work to determine the QRF is

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based,

$$\begin{aligned}
& - \frac{\text{Re}\hat{\chi}_{ijk}^{(2)}(\mathbf{k}_1, \omega_1; \mathbf{k}_2, \omega_2)}{\omega_1\omega_2} - \frac{\text{Re}\hat{\chi}_{jki}^{(2)}(\mathbf{k}_2, \omega_2; \mathbf{k}, \omega)}{\omega\omega_2} \\
& - \frac{\text{Re}\hat{\chi}_{kij}^{(2)}(\mathbf{k}, \omega; \mathbf{k}_1, \omega_1)}{\omega_1\omega} = \frac{-n}{4\hbar^2} \left\{ \frac{S(102) + S(201)}{\omega_1\omega_2} \right. \\
& \left. - \frac{S(012) + S(210)}{\omega\omega_2} - \frac{S(021) + S(120)}{\omega_1\omega} \right\}. \quad (1)
\end{aligned}$$

Here, in the left part, there is a combination of the real parts of the Fourier images of tensor QRF $\hat{\chi}_{ijk}^{(2)}$, which determine the quadratic contribution to the polarization of $\mathbf{P}^{(2)}$ ($\mathbf{P} = \sum_i e_i \mathbf{r}_i / V$; e_i , $\mathbf{r}_i$, V represents the charge of the particle, its radius vector and the volume of the system; n is the particle concentration, $\hbar$ is Planck's constant) of a charged medium under the action of an external electromagnetic field. Accordingly, in the right part of (1) there is a combination of the polarization correlators (indicated by B_i),

$$\begin{aligned}
& \langle B_i(-\mathbf{k}, -\omega) B_j(\mathbf{k}_1, \omega_1) B_k(\mathbf{k}_2, \omega_2) \rangle_0 = \\
& 2\pi N \delta(\omega - \omega_1 - \omega_2) \delta(\mathbf{k} - \mathbf{k}_1 - \mathbf{k}_2) S(012). \quad (2)
\end{aligned}$$

In the correlator, the set $\{a, b, c\}$ means a combination of the wave vector and frequency $\{k, \omega\}$ with the corresponding indices, $\langle \dots \rangle_0$ means averaging over the equilibrium density matrix. A quadratic FDT corresponding to QRF (denoted by $\hat{\chi}_{J,ijk}^{(2)}$) looks similar, describing the quadratic contribution of $\mathbf{j}^{(2)}(k, \omega)$ to the electric current density $\mathbf{J}$ of a charged medium under the action of an electromagnetic field.

Expressions (1) and (2) are applicable for calculating the frequency moments of the real part of the QRF, since they relate the frequency moments of the reaction functions to the frequency moments of the correlators. Note that in the linear case, the correlator corresponds to the imaginary part of the reaction function, and the FDT contains one reaction function and one correlator (see, for example, [4]).

The quadratic FDT looks the most simple in the classical limit, because in this case the correlators are invariant with respect to the permutation of arguments. Laboratory dense plasma is often nondegenerate and obeys classical statistics [1–4]. At the same time, the nonlinear FDT in the form (1), (2) is inconvenient when implementing a model approach consisting in applying explicit approximations for the response functions $\hat{\chi}_{ijk}^{(2)}$ with fitting parameters and exact values of the frequency moments of the corresponding correlators, since the tensors in (1) have 18 components each, which are not invariant with respect to maps. Therefore, in the classical limit, by performing a longitudinal projection of the tensors in (1), we obtain a nonlinear FDT in the form of an invariant relation for scalar QRF $\hat{\chi}^{(2)}$ plasma in a constant magnetic field to the longitudinal field ($\hat{\chi}^{(2)}$ stands for the contribution $\rho^{(2)}(k, \omega)$ to the total charge density ρ) and the correlator of charge densities (for more information, see [2,3]). In

this case, we write out the longitudinal quadratic FDT,

$$\begin{aligned}
& - \omega \text{Re}\hat{\chi}^{(2)}(12) - \omega_1 \text{Re}\hat{\chi}^{(2)}(20) - \omega_2 \text{Re}\hat{\chi}^{(2)}(01) = \\
& \frac{n\beta^2}{4} S_\rho(012) \omega \omega_1 \omega_2. \quad (3)
\end{aligned}$$

In (3) $\hat{\chi}^{(2)}$ is the QRF of the charge density [2,3] on the external field, the set $\{a, b\}$ means a combination k, ω with the corresponding indices, $\beta = \frac{1}{k_B T}$, k_B is the Boltzmann constant, T is the temperature. By setting the QRF, according to the proposed model (see above and eqs. (4), (5)), in the form of an explicit approximation with fitting parameters, these parameters can be determined by comparing the frequency moments of the right and left parts of eq. (3).

Using (3) and the results for the frequency moments $S_\rho(012)$ (taken from [2,3]), we can find “scalar” fitting parameters for $\hat{\chi}^{(2)}$, depending on the thermophysical characteristics of the charged medium (dense plasma, Coulomb system) and the magnetic field. It is advisable to use these parameters for the entire line of interconnected QRFs ($\hat{\chi}_{ijk}^{(2)}$, $\hat{\chi}_{J,ijk}^{(2)}$, $\hat{\chi}^{(2)}$).

Simulation of the quadratic response function.

– Since, as mentioned above, neither a theoretical nor a numerical analysis of the specific form of $\hat{\chi}^{(2)}$ (or related functions) has been performed for media with strong interparticle interaction, it is natural to begin the study of QRF with one-parameter models. Let us choose an approximation for $\hat{\chi}^{(2)}$ (the sign Σ means summation by charge grades, but the main contribution to $\hat{\chi}^{(2)}$ is made by electrons, see (6)) in the form of (4) and (5), since this form has the necessary tensor structure, it is comparable with the asymptotics of QRF for cold rarefied plasma in a magnetic field, and is convenient for calculations. Below, δ_{ij} is a single symmetric tensor, $b_i = b_z$ is a single vector in the direction of the magnetic field, ε_{ijk} is a completely antisymmetric tensor, ω_B and ω_p are cyclotron and plasma frequencies, and “ ν ” is the desired parameter,

$$\begin{aligned}
& \hat{\chi}^{(2)}(12) = \sum \frac{(-i)}{2} \frac{e}{m} \frac{\omega_p^2}{\omega \omega_1 \omega_2} \left(\frac{-i}{4\pi} \right)^3 \\
& \times \{ \omega (k\Gamma(\omega_1)k_1)(k\Gamma(\omega_2)k_2) - \omega_1 (k\Gamma(\omega)k_1)(k_1\Gamma(\omega_2)k_2) \\
& - \omega_2 (k\Gamma(\omega)k_2)(k_2\Gamma(\omega_1)k_1) \}, (k\Gamma(\omega_1)k_1) \equiv k_i \Gamma_{ij}(\omega_1) k_{1j}, \quad (4) \\
& \omega_p^2 \Gamma_{ij}(\omega_1) \Gamma_{lk}(\omega_2) = \left\{ \frac{1}{2} [e^{-(\omega_1-\nu)^2/\omega_p^2} + e^{-(\omega_1+\nu)^2/\omega_p^2}] \right. \\
& \times (\delta_{ij} - b_i b_j) - \frac{i}{2} [e^{-(\omega_1-\nu)^2/\omega_p^2} - e^{-(\omega_1+\nu)^2/\omega_p^2}] \varepsilon_{ijf} b_f \\
& + e^{-(\omega_1+\nu)^2/\omega_p^2} b_i b_j \left. \right\} \frac{\omega}{\omega_p} \left\{ \frac{1}{2} [e^{-(\omega_2-\nu)^2/\omega_p^2} + e^{-(\omega_2+\nu)^2/\omega_p^2}] \right. \\
& \times (\delta_{lk} - b_l b_k) - \frac{i}{2} [e^{-(\omega_2-\nu)^2/\omega_p^2} - e^{-(\omega_2+\nu)^2/\omega_p^2}] \varepsilon_{lkm} b_m \\
& + e^{-(\omega_2+\nu)^2/\omega_p^2} b_l b_k \left. \right\} \frac{\omega_1}{\omega_p} \frac{\omega_2}{\omega_p}. \quad (5)
\end{aligned}$$

Next, multiply the right and left sides of (3) by $\omega_1 \omega_2$, divide by ω ; set the set of wave vectors in the form

Table 1: PGR parameters for laboratory plasma (see (11), (12)). $k \equiv k_z, k_z = 2.3\omega_p/c$, $k_{1x} = -k_{2x} = 10^{-1}\omega_p/c$, $k_{1z} = 1.15\omega_p/c = k_{2z}$; $f = a(0)\text{Re}\sigma_1/\Delta k$, $\omega_1 = \omega_2 = \omega/2$, $r_s = (a(0)\text{Re}\sigma_1)^{-1}$, $s = a(0)\text{Re}\sigma_1 - \delta_1$.

B (G)	T (eV)	n_e (10^{21} cm^{-3})	ν, ω_p	$a(0)$ (V/cm)	ω_1, ω_p	δ_1 (cm^{-1})	$\text{Re}\sigma_1$ (V^{-1})	s (cm^{-1})	
								$f = \infty$	$f = 2$
10^4	3	1	$-1/3$	10^5	1.6	$2.4 \cdot 10^2$	10^{-5}	$s < 0$	$s < 0$
10^4	3	10^{-3}	-0.28	10^4	1.6	0.25	$2 \cdot 10^{-5}$	$s < 0$	$s < 0$
10^5	3	10^{-5}	0.26	10^4	1.6	0.003	10^{-4}	$r_s \sim 1$	$\tilde{r}_s \sim 1$

$\{k = k^z; k_1 = k_1^x, k_1^z; k_2 = k_2^x, k_2^z\}$. The left part of (3) ($-\omega_1\omega_2[\omega\text{Re}\hat{\chi}^{(2)}(12)+\omega_1\text{Re}\hat{\chi}^{(2)}(20)+\omega_2\text{Re}\hat{\chi}^{(2)}(01)]$; see (4), (5)) we integrate by the frequencies ω_1 and ω_2 . As a result, we find the moment of the sum of the real parts $\hat{\chi}^{(2)}$ with the corresponding arguments containing the desired parameter “ ν ” (with electronic cyclotron and plasma frequencies ω_B, ω_p),

$$\begin{aligned} & \int_{-\infty}^{\infty} d\omega_1 d\omega_2 \omega_1 \omega_2 [\omega \text{Re}\hat{\chi}^{(2)}(12) + \omega_1 \text{Re}\hat{\chi}^{(2)}(20) \\ & + \omega_2 \text{Re}\hat{\chi}^{(2)}(01)] = \frac{(-i)}{2} \left(\frac{-i}{4\pi} \right)^3 \frac{e\omega_p^2}{m_e} [2\pi k^z k_1^x k_2^x (1 + 2q^2) \\ & \times (k_1^z + k_2^z) - \pi k^z k^z k_1^z k_2^z q (1 + 2q^2)]. \end{aligned} \quad (6)$$

This integral corresponds to the moment of the correlator $S_{\rho}^{2,2} = \int_{-\infty}^{\infty} d\omega_1 d\omega_2 \omega_1^2 \omega_2^2 S_{\rho}(012)$, calculated in [2] ($q = \nu/\omega_p, \omega_B = eH/cm_e; e, m_e, H, c$ are the charge and mass of the electron, the magnetic field and the speed of light; $\omega_p^2 = 4\pi n_e e^2/m_e$). Equating (6) (with the opposite sign) and $\frac{n\beta^2}{4} S_{\rho}^{2,2}$ (see (3)), we find that the parameter “ ν ” is determined from the following equation:

$$\begin{aligned} & \frac{1}{2} \left(\frac{1}{4\pi} \right)^3 \frac{e\omega_p^2}{m_e} 2\pi k^z k^z k_1^z k_2^z \left[q^3 - \left(\frac{k_1^x k_2^x}{k^z k_1^z} + \frac{k_1^x k_2^x}{k^z k_2^z} \right) q^2 \right. \\ & \left. + \frac{1}{2} q - \frac{1}{2} \left(\frac{k_1^x k_2^x}{k^z k_1^z} + \frac{k_1^x k_2^x}{k^z k_2^z} \right) \right] - \frac{n\beta^2}{4} S_{\rho}^{2,2} = 0. \end{aligned} \quad (7)$$

The paper considers plasma in a permanent magnetic field, therefore, we investigate the case of a significant effect of this field on $\hat{\chi}^{(2)}$ (magnetized plasma) when the contribution to $S_{\rho}^{2,2}$ of the corresponding terms is decisive (i.e., plasma when the parameter $\alpha = \frac{k^2}{\beta}/\frac{H^2 e^2}{m_e c^2}$ [2] becomes small compared to 1, see below). Let us write out $S_{\rho}^{2,2}$ in this case,

$$\begin{aligned} S_{\rho}^{2,2} \approx & - \left[\frac{H^2}{V} k_1^x k_2^x \frac{e^5}{c^2} \frac{1}{\beta m_e^3} \delta_{\mathbf{k}+\mathbf{k}_1+\mathbf{k}_2} + \frac{H^2 N'}{V^2} k_1^x k_2^x \frac{e^5}{c^2} \frac{1}{\beta m_e^3} \right. \\ & \left. \times \int e^{-i(\mathbf{k} \cdot \mathbf{r} + (\mathbf{k}_1 + \mathbf{k}_2) \cdot \mathbf{r}_1)} g_{ee}(r, r_1) d\mathbf{r} d\mathbf{r}_1 \right]. \end{aligned} \quad (8)$$

Leaving in (8) the first (maximum) term on the right side, we obtain a third-degree equation for determining the

desired parameter “ ν ” by q_+ (roots of (9) [5], see table 1),

$$\begin{aligned} & q^3 - \left(\frac{k_1^x k_2^x}{k^z k_1^z} + \frac{k_1^x k_2^x}{k^z k_2^z} \right) q^2 + \frac{1}{2} q - \frac{1}{2} \left(\frac{k_1^x k_2^x}{k^z k_1^z} + \frac{k_1^x k_2^x}{k^z k_2^z} \right) \\ & - \frac{n\beta^2}{4} S_{\rho}^{2,2} / \frac{1}{2} \left(\frac{1}{4\pi} \right)^3 \frac{e\omega_p^2}{m_e} 2\pi k^z k^z k_1^z k_2^z = 0, \end{aligned} \quad (9)$$

$$q \equiv \frac{\nu}{\omega_p}.$$

Note that if data on computer modeling of QRF may appear in the future, an approximation of the results of such modeling $\text{Re}\hat{\chi}^{(2)}$ and related QRF can be used to simplify calculations in the form of a one-parameter sum of exponentials with a fitting parameter, which is determined from an algebraic equation of type (9).

Then we write out the QRF $\hat{\chi}_{ijk}^{(2)}$ taking into account all kinds of charges in the medium (the longitudinal projection of this reaction function is $\hat{\chi}^{(2)}$ in the form (4)),

$$\begin{aligned} \hat{\chi}_{ijk}^{(2)}(1, 2) = & \sum \frac{(-i)}{2} \frac{e}{m} \frac{\omega_p^2}{\omega \omega_1 \omega_2} [\omega_1 \Gamma_{ij}(\omega_1) \Gamma_{lk}(\omega_2) k_l \\ & + \omega_2 \Gamma_{ik}(\omega_2) \Gamma_{lj}(\omega_1) k_l + \omega_1 \Gamma_{kj}(\omega_1) \Gamma_{il}(\omega) k_{2l} \\ & - \omega \Gamma_{ik}(\omega) \Gamma_{lj}(\omega_1) k_{2l} + \omega_2 \Gamma_{jk}(\omega_2) \Gamma_{il}(\omega) k_{1l} \\ & - \omega \Gamma_{ij}(\omega) \Gamma_{lk}(\omega_2) k_{1l}]. \end{aligned} \quad (10)$$

This expression (taking into account (5) and the data on “ ν ”) can be used in the study of PGR and SHG phenomena in dense laboratory plasma, Coulomb systems and other charged media with classical statistics. At the same time, we clarify that $\hat{\chi}_{ijk}^{(2)}$ is a quadratic response function to an external field $D(r, t)$ [2]. At the same time, when studying nonlinear phenomena in dense plasma, one should use the equation with respect to the polarization of the medium P and the electric field $E(r, t)$ (see, for example, [1, 6–9]), in which the QRF for the average field in a given medium appears (let us denote them $\chi_{ijk}^{(2)}$). Therefore, in order to construct a model for $\chi_{ijk}^{(2)}$ by $\hat{\chi}_{ijk}^{(2)}$, a refinement is necessarily related to the substitution of the well-known ratio $\mathbf{D} = \hat{\epsilon} \mathbf{E}$ ($\hat{\epsilon}$ is the dielectric constant of the medium) in the definition of $\mathbf{P}^{(2)}$ through $\chi_{ijk}^{(2)}$ and $\mathbf{E}$: $\mathbf{P}_i^{(2)} = \chi_{ijk}^{(2)} E_j E_k$ (and not through $\hat{\chi}_{ijk}^{(2)}$ and $\mathbf{D}$) into the equations describing the PGR and SHG.

Nonlinear phenomena of PGR and SHG in dense plasma. – Let us move on to comparing the characteristics of PGR and SHG obtained in [1] (see the table

in [1]), where the fitting parameter, an analog of the parameter “ ν ”, is set without taking into account the external magnetic field and the approximation $\chi_{ijk}^{(2)}$ is used [1,7,8], differently from the one used above (the form $\hat{\chi}_{ijk}^{(2)}$, see (4), (5), (10)). In this work, the characteristics of PGR (see table 1) and SHG were obtained within the framework of a new model (see previous section), where the parameter “ ν ” is consistently determined taking into account the external magnetic field, the shape of $\hat{\chi}_{ijk}^{(2)}$, selected according to 10, and modified taking into account the correction $\mathbf{D} = \hat{\varepsilon}\mathbf{E}$. In this case, the condition $\text{Re}\chi_{ijk}^{(2)} > \text{Im}\chi_{ijk}^{(2)}$ is fulfilled.

Let us write down the equations describing the PGR phenomenon in the medium (see [1,6]). When $\chi_{ijk}^{(2)} \approx \text{Re}\chi_{ijk}^{(2)}$ the field of “pumping waves” (PW) with amplitude a and frequency ω is set. Let us assume that the amplitude a decreases just due to the absorption of δ in the medium, and not due to the pumping of energy into the “signal” SW wave (amplitude a_1 and frequency ω_1), the “idle” IW wave (amplitude a_2 and frequency ω_2). In addition, the condition holds: $a_{1,2}(z) \ll a(z=0)$ (see notation (12)). Let $\delta n = \delta$, $\omega_1 = \omega_2$, $\omega = 2\omega_1$, $\sigma_1 = \sigma_2$, $a_1 = a_2$, and $\Psi(z) \approx \pi/2 + \Delta\mathbf{k} \cdot \mathbf{z}$ be the equation for the phase. After these minor simplifications, the equation for a_1 will take the form (see for more details [1,6])

$$a_1(z) = a_1(0) \exp \left\{ \left[\text{Re}\sigma_1 a(0) \frac{\sin(\Delta k z)}{\Delta k z} - \delta_1 \right] z \right\}. \quad (11)$$

In (11), which is convenient for analyzing the conditions of the PGR implementation in the medium, the notation (12) is introduced, $\Delta\mathbf{k} = \mathbf{k} - \mathbf{k}_1 - \mathbf{k}_2$, $\Delta\mathbf{k}$ is the projection $\Delta\mathbf{k}$ on the z -axis, $n(\omega)$ is the refractive index at the corresponding frequency, $A_n(z) = a_n(z)e^{i\varphi_n(z)}$ is the complex amplitude,

$$\begin{aligned} \Psi &\equiv \varphi - \varphi_1 - \varphi_2 - \Delta\mathbf{k} \cdot \mathbf{z}; \\ \delta_n &= \omega_n \mathbf{e}_n \frac{\text{Im}\varepsilon(\omega_n) \mathbf{e}_n}{2n(\omega_n)c}; \quad k = \omega n(\omega)/c; \\ \sigma_1 &= 2\pi\omega_1 \mathbf{e}_1 \frac{\chi^{(2)}(\omega_1): \mathbf{e}_2 \mathbf{e}_1}{n(\omega_1)c}; \\ \sigma_2 &= 2\pi\omega_2 \mathbf{e}_2 \frac{\chi^{(2)}(\omega_1): \mathbf{e} \mathbf{e}_1}{n(\omega_2)c}; \\ \sigma &= 2\pi\omega \mathbf{e} \frac{\chi^{(2)}(\omega): \mathbf{e}_1 \mathbf{e}_2}{n(\omega)c}, \\ \varepsilon_{ij} &= \delta_{ij} + i \frac{4\pi}{\omega} \sigma_{ij}(\mathbf{k}, \omega); \\ \sigma_{ij}(\mathbf{k}, \omega) &= \begin{Bmatrix} \sigma_{xx} & \sigma_{xy} & 0 \\ \sigma_{yx} & \sigma_{yy} & 0 \\ 0 & 0 & \sigma_{zz} \end{Bmatrix}; \\ \sigma_{xx} = \sigma_{yy} &= \frac{\omega_p^2}{4\pi} \frac{i\omega + \tau^{-1}}{(i\omega + \tau^{-1})^2 + \omega_B^2}, \\ \sigma_{xy} = -\sigma_{yx} &= \frac{\omega_p^2}{4\pi} \frac{\omega_B}{(i\omega + \tau^{-1})^2 + \omega_B^2}, \end{aligned} \quad (12)$$

$$\sigma_{zz} = \frac{\omega_p^2}{4\pi} \frac{1}{(i\omega + \tau^{-1})^2}.$$

In (12), δ_n and σ_n are convolutions of tensors ε_{ij} and $\chi_{ijk}^{(2)}$ with polarization vectors at the corresponding indices, $\mathbf{e}_i$ are the unit single polarization vectors of EM waves; τ^{-1} is the “collision frequency” (a parameter for calculations taken from [4]). There follows from (11) that the conditions for exponential growth of the SW amplitude are the positivity of the real part σ_1 and $\text{Re}\sigma_1 a(0) > \delta_1$ [1] (and $\Delta\mathbf{k} \cdot \mathbf{z} = 0$). An increase in the SW amplitude in z is also possible when a more complete system of equations for amplitudes and $\text{Re}\sigma_1 a(0) > \Delta k$ ($f = 2$, see table 1) is used instead of (11). The fulfillment and specification of the listed conditions depend on the intensity and polarization of the interacting electromagnetic waves (convolutions in definitions (12) should not be zero), the type and values of the real and imaginary parts of the tensor, H and the linear properties of the medium (δ_1 , see table 1). Note that the linear dynamic properties, ε_{ij} , of a dense charged medium are quite well studied (see, for example, [4]), but the nonlinear reaction of such a medium to an electromagnetic field was studied only in [1]. Parameters $\{\sigma_n\}$ in isotropic plasma, are equal to zero.

In the case of SHG, by one wave of the frequency ω_1 with an amplitude a_1 , can be obtained for the amplitude of the second harmonic a_2 (by analogy with (11), (12)): $a_2(z) \approx \sigma_2 a_1^2 z$ (on the coherence length, if $\sigma_i \approx \text{Re}\sigma_i$, $\sigma_2 a_1^2 \cos(\Delta k z) - \delta_2 a_2 > 0$).

To compare the characteristics of the PHI according to various models (data from [1] and from table 1), we calculate, within the framework of a self-consistent model (see the two previous sections), for the parameters of electromagnetic waves and laboratory plasma (*i.e.*, for a fully ionized hydrogen plasma) δ_1 , the tensor $\chi_{ijk}^{(2)}$, $\text{Re}\sigma_1$, ν/ω_p (see table 1). To evaluate the parameters $\{\sigma_n\}$ in (11), (12), the configuration $\{\mathbf{k}_i\}$ and $\{\mathbf{e}_i\}$ is set as follows: $\mathbf{k} \parallel \mathbf{H}$, $\{\mathbf{k}_i\}$ are in the same plane, $\{\mathbf{e}_i\}$ have components $\mathbf{e}_y, \mathbf{e}_{1,x}, \mathbf{e}_{2,x}$. According to the type of tensor $\chi_{ijk}^{(2)}$ and the values of the parameter ν , it can be concluded that in the proposed model $\chi_{ijk}^{(2)} \approx \text{Re}\chi_{ijk}^{(2)}$. In this case, the values of parameters σ_1 and δ (12) for the plasma are found using the known tensors $\chi_{ijk}^{(2)}$, ε_{ij} and $\{\mathbf{k}_i\}$, $\{\mathbf{e}_i\}$. These results make it possible, for a given intensity of PW, to estimate the characteristic length of the effective power transmission from PW to SW, at which the amplitude of the SW increases e -fold based on the self-consistent model proposed in the work (see the previous section) ($r_s, \tilde{r}_s$ at $\Delta k = 0$ and $\Delta k \neq 0$, respectively). The dependence of the parameters $\{\sigma_n\}$ on $\{\mathbf{k}_i\}$, $\{\mathbf{e}_i\}$ is conservative. This circumstance makes it possible to carry out a fairly general assessment of the conditions for the implementation of PGR for laboratory dense plasma with a given approximation of $\hat{\chi}_{ijk}^{(2)}$ (see (4), (5), (10)).

Table 1 shows the values of the PGR parameters for fully ionized hydrogen plasma of various concentrations,

estimated according to the model proposed in the previous section. Thermodynamic functions (pressure, enthalpy, etc.) corresponding to the parameters of a fully ionized hydrogen plasma with a strong Coulomb interaction from table 1 can be determined using techniques and tabular data from [10].

Note that the condition $f = \infty$ (the condition of phase synchronism [6]) in plasma cannot be precisely fulfilled due to the sharp dependence of the plasma refractive index on frequency. Nevertheless, the use of (11), (12) at $\Delta k/k$ small compared to 1 for a qualitative discussion of the conditions for the occurrence of PGR in laboratory plasma is acceptable (cf. $r_s, \tilde{r}_s$ at $f = \infty, 2$ in table 1). An analysis of the obtained parameter values shows that the quadratic interaction of electromagnetic waves in a spatially homogeneous plasma in an external constant and homogeneous magnetic field at experimentally achievable values of the magnetic field and $a(0)$ (such fields do not provoke “heating” and “friction” instabilities [11], see table 1) leads to the realization of the PGR phenomenon at $r_s \sim \text{cm}$ only for sufficiently moderate concentrations of charged particles. This circumstance is due both to the very small values of the nonlinear reaction functions and to the high level of radiation absorption in a denser plasma. In other words, in this model and the model from [1], a PGR “cutoff” is realized in a nonideal plasma, but for the self-consistent model (see the previous section), this occurs at higher concentrations of charged particles for a given intensity of HF and vice versa.

In the case of SHG, the “cutoff” corresponds to a violation of the condition $\sigma_2 a_1^2 \cos(\Delta k z) - \delta_2 a_2 > 0$ (see above) in a strongly or fully ionized dense plasma. As the analysis shows, in order to overcome the “cutoff”, it is necessary to increase the intensity of exposure to the environment, which can lead to a different type of reaction of the environment (see, for example, [12]).

Conclusion. – Thus, for the first time, a consistent technique for constructing a quadratic response function to the electromagnetic field of a dense plasma (Coulomb system) with classical statistics in a constant magnetic field was proposed and implemented (see section “Simulation of the quadratic response function”). The technique is based on the use of the quadratic fluctuation-dissipation theorem, known frequency moments for the QRF of a charged medium, and an explicit approximation for the QRF with fitting parameters. It is shown that in this model (as in the model from [1]), a “cutoff” of PGR is realized in a nonideal plasma, but for a self-consistent model (see section “Simulation of the quadratic response function”) this occurs at higher concentrations of charged particles for a given pump wave intensity than in [1].

Note that the approximation for the response function can be chosen in various forms with a tensor structure satisfying the general properties of symmetry and reality of the QFR (see, for example, [3,8]). In practical

implementation, the proposed technique based on comparing the frequency moments of the right and left sides of FDT (3) becomes rather cumbersome when the contribution of the magnetic field to the correlators (for example, in $S_{\rho}^{2,2}$) is not decisive (parameter $\alpha \sim 1$ and all terms in $S_{\rho}^{2,2}$ [2,3] must be taken into account) or an approximation for QRF is selected as multiparametric one (in this case, the required number of moments S_{ρ} should be used). In order to optimize the above-described method for determining QRF, in particular, when choosing an explicit approximation of this function, it will be necessary to take into account the results of computer modeling of nonlinear response functions for the simplest Coulomb systems, as well as data from future experimental studies of PGR, SHG and other nonlinear phenomena in dense charged media with high-energy concentrations.

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